Research Paper

Study on Measuring Pedagogical Alignment in Online Learning Applications Using Evidence-Based Learning and Bloom's Taxonomy

Project Category: Research Based Project

Projexa Team Id- 26E3257

Submitted in partial fulfilment of the requirement of the degree of

BCA Specialization in (AI & DS) (Section -A)

to

K.R Mangalam University

by

Piyush Vats – 2401201025

Karamveer Singh – 2401201072

Himanshi Tanwar – 2401201053

Yashika Gour – 2401201012

Under the supervision of :

Mr. Shaquib Hasan

Department of Computer Science and Engineering

School of Engineering and Technology

K.R Mangalam University, Gurugram- 122001, India

INDEX

ABSTRACT

In the modern digital era, online learning applications have become an essential part of education, especially for young learners in preschool and foundational stages. These applications are widely used due to their easy accessibility, interactive design, and engaging user interface, making them attractive to both students and parents. However, despite their increasing popularity, there is a growing concern regarding their actual educational effectiveness.

Most learning applications are designed with a strong focus on engagement techniques such as animations, rewards, gamification, and sound effects. While these features successfully capture user attention, they do not necessarily ensure deep understanding or cognitive development. This creates a major gap between perceived learning (what users think they learn) and actual learning (what they truly understand and apply).

Another major issue is the lack of a standardized evaluation system to measure whether these applications follow proper educational principles. As a result, parents and educators often rely on ratings and popularity, which do not accurately reflect learning quality.

This research aims to address this problem by evaluating online learning applications using **Bloom's Taxonomy** and **Evidence-Based Learning (EBL) principles**. A structured framework is developed to measure important factors such as:

- Active engagement
- Meaningful learning
- Cognitive level
- User interaction

The study involves analyzing multiple learning apps and assigning scores based on these criteria. The outcome is a Pedagogical Evaluation Framework that helps in identifying whether an app supports effective learning or just entertainment.

INTRODUCTION

Technology has significantly transformed the field of education in recent years. Online learning applications have become one of the most popular tools for delivering education, especially among children. These apps provide learning through games, videos, quizzes, and interactive activities, making the learning process more engaging and enjoyable.

Today, many parents prefer using learning apps because they are convenient, flexible, and easily accessible anytime and anywhere. These applications claim to improve essential skills such as:

- Reading
- Writing
- Logical thinking
- Problem-solving

However, despite these claims, there is a major concern regarding their actual learning impact.
Most applications are designed to maximize user engagement rather than learning outcomes. They use visual elements, rewards, and gamification to keep users interested. While this increases app usage, it does not guarantee concept clarity or deep understanding.

Many apps focus only on:

- **Memorization**
- **Repetition**

Instead of:

- **Understanding**
- **Application**
- **Critical thinking**

This creates a situation where learners appear to be learning but are not actually developing higher-order cognitive skills.
To address this issue, it is important to evaluate learning apps using scientific and educational frameworks such as:

- **Bloom’s Taxonomy** (for cognitive levels)
- **Evidence-Based Learning** (for engagement and effectiveness)

MOTIVATION

The motivation behind this research comes from the rapid increase in the use of online learning applications, especially among young children. With the growth of digital education, children are spending more time on mobile devices for learning purposes.

Parents and educators trust these apps because they are marketed as educational tools that improve learning outcomes. However, during initial observation and analysis, it was found that many applications prioritize:

“Fun over learning”
They use:

- Bright visuals
- Sounds
- Rewards
- Gamification

These features attract attention but do not guarantee understanding.
This creates a serious issue where:

- Children spend more time on apps
- But gain less meaningful knowledge

Another important concern is that:
App ratings and reviews do NOT reflect educational quality

They mostly represent:

- User satisfaction
- Interface design
- Entertainment value

Not:

- Learning effectiveness

Therefore, there is a strong need to identify what makes an app truly educational

This project is motivated by the idea of:

- Creating a scientific evaluation system
- Helping users choose better learning tools
- Encouraging developers to design learning-focused apps

LITERATURE REVIEW:

The field of online learning and educational technology has seen significant growth in recent years, especially with the increasing use of mobile applications for early education. Many researchers have studied how digital platforms can improve learning experiences and what factors contribute to effective education through technology. These studies highlight that while online learning applications have the potential to transform education, their effectiveness largely depends on how well they are designed according to established educational principles.

One of the most widely accepted frameworks in the field of education is **Bloom's Taxonomy**, which provides a structured way to understand different levels of learning. According to this framework, learning is not limited to simple memorization but progresses through multiple stages such as **understanding, applying, analyzing, evaluating, and creating**. Research studies emphasize that effective learning applications should not only focus on basic knowledge recall but should also encourage **higher-order thinking skills**, which are essential for long-term learning and problem-solving abilities. However, several studies have pointed out that many existing learning applications primarily focus on lower levels of learning, such as remembering and basic understanding, while neglecting advanced cognitive skills.

Another important concept discussed in the literature is **Evidence-Based Learning (EBL)**, which focuses on improving learning outcomes by using scientifically proven teaching methods. EBL highlights the importance of **active engagement, meaningful learning, and interaction** in the learning process. Research suggests that learners understand concepts better when they are actively involved in the learning activity rather than passively consuming information. This means that effective learning applications should include features such as interactive tasks, feedback systems, and real-life problem-solving activities. However, many modern applications fail to implement these principles effectively and instead rely on superficial engagement techniques.

Several studies also highlight a growing trend in the design of online learning applications, where developers prioritize **user engagement over educational effectiveness**. Features such as animations, rewards, badges, and gamification are commonly used to attract and retain users. While these features increase the time users spend on the application, they do not necessarily contribute to meaningful learning. Research has shown that excessive focus on entertainment can lead to **passive learning**, where users interact with the application without gaining a deep understanding of the subject matter.

In addition to this, existing research points out the **lack of proper evaluation systems** for measuring the educational quality of learning applications. Most applications are judged based on user ratings, reviews, and popularity, which primarily reflect user experience rather than actual learning outcomes. This creates a challenge for parents and educators, as they are unable to differentiate between applications that are truly educational and those that are mainly designed for entertainment purposes. Furthermore, studies in this field suggest that there is a need for a **standardized framework** that can evaluate learning applications based on pedagogical principles. Such a framework should consider factors like cognitive development, user interaction, and learning effectiveness. Without such a system, it becomes difficult to ensure that digital learning tools are contributing positively to the educational development of learners.

Overall, the literature clearly indicates that while online learning applications have great potential, there is a significant gap between their intended purpose and actual performance. Many applications fail to align with established educational theories, resulting in limited learning outcomes. Therefore, there is a strong need for research that focuses on developing a structured and scientific approach to evaluate the pedagogical alignment of learning applications and ensure that they support meaningful and effective learning.

GAP ANALYSIS

After analyzing existing research and current online learning applications, several important gaps have been identified in the field of digital education. Although many studies focus on improving user engagement and application usability, there is still a lack of emphasis on **actual learning effectiveness and cognitive development**. Most existing research evaluates applications based on how attractive or user-friendly they are, but does not properly measure whether they help learners understand concepts in a meaningful way.

One of the major gaps identified is the **absence of a standardized evaluation framework** for learning applications. Currently, there is no common system or method that can be used to assess whether an application follows proper educational principles. As a result, different apps cannot be compared on a uniform basis, making it difficult to identify which apps are truly effective for learning and which are not. This lack of standardization creates confusion for both parents and educators when selecting suitable learning tools.

Another important gap is related to the **limited focus on higher-order thinking skills**. Most existing applications are designed to support basic learning activities such as memorization and repetition. While these are important for initial learning, they are not sufficient for developing deeper understanding. Research shows that effective learning requires skills such as **analysis, problem-solving, and critical thinking**, but these are often missing in many digital learning platforms.

In addition to this, there is a noticeable gap in the implementation of **Evidence-Based Learning principles**. Many applications do not provide opportunities for active engagement or meaningful interaction. Instead, they rely heavily on passive content delivery, where users simply watch or click without fully engaging with the material. This reduces the effectiveness of the learning process and limits knowledge retention.

Another key issue is the **over-dependence on engagement-based features** such as rewards, animations, and gamification. While these features make the application more attractive, they often distract from the actual learning objectives. This results in a situation where users are more focused on earning rewards rather than understanding concepts.

Furthermore, there is a lack of **quantitative scoring systems** that can measure and compare the educational quality of different applications. Without a proper scoring mechanism, it becomes difficult to evaluate the strengths and weaknesses of each app in a structured way.

Therefore, this research aims to fill these gaps by developing a **structured Pedagogical Evaluation Framework** that focuses on **learning effectiveness, cognitive development, and educational alignment**, rather than just engagement.

PROBLEM STATEMENT

With the rapid advancement of digital technology, online learning applications have become an integral part of early education. These applications are widely used by children, parents, and educators due to their **easy accessibility, flexibility, and interactive design**. However, despite their growing popularity, there is a major concern regarding their **actual educational effectiveness and learning outcomes**.

Most online learning applications are designed with a strong focus on **user engagement rather than educational value**. They use various techniques such as animations, sound effects, rewards, and gamification to attract users and keep them engaged for longer periods. While these features are effective in increasing user interaction, they do not necessarily contribute to **deep understanding or meaningful learning**. As a result, learners may spend a significant amount of time using these applications without gaining proper knowledge or conceptual clarity.

Another critical issue is that many learning applications do not follow **established educational frameworks or pedagogical theories**. They mainly focus on basic learning activities such as memorization and repetition, while ignoring higher-level cognitive processes like **application, analysis, and problem-solving**. This limits the overall development of learners and reduces the effectiveness of these applications as educational tools.

Additionally, there is a **lack of a standardized system** to evaluate the quality of learning applications. Most users rely on ratings, reviews, and popularity when selecting apps, but these factors do not accurately reflect the **educational quality or learning effectiveness** of the application. This creates confusion and makes it difficult to identify which apps are truly beneficial for learning.

Another important problem is the **gap between perceived learning and actual learning**. Many users believe that they are learning effectively because the app is engaging and interactive, but in reality, they may not be developing essential cognitive skills. This leads to a false sense of learning and reduces the overall impact of digital education.

Therefore, there is a strong need to develop a scientific and structured approach to evaluate online learning applications based on pedagogical alignment, cognitive development, and learning effectiveness, ensuring that these platforms truly support meaningful education.

OBJECTIVES:

The main objective of this research is to analyze and evaluate online learning applications in order to determine whether they provide **real educational value or simply focus on user engagement**. This study aims to bridge the gap between digital interaction and actual learning by applying established educational frameworks.

One of the primary objectives is to study the current landscape of online learning applications, especially those designed for **preschool and foundational level learners**. This includes understanding their features, design patterns, and types of learning activities they provide. By analyzing these aspects, the study aims to identify common trends and limitations in existing applications.

Another important objective is to evaluate these applications using **Bloom's Taxonomy**, which helps in identifying the level of cognitive skills involved in the learning process. This includes analyzing whether the app supports basic levels such as remembering and understanding, or higher levels such as **application, analysis, and problem-solving**. The goal is to determine whether these apps encourage deeper thinking or just surface-level learning.

The study also aims to analyze applications based on **Evidence-Based Learning (EBL) principles**, which focus on improving learning outcomes through active engagement and meaningful interaction. This involves evaluating whether the applications provide opportunities for users to actively participate, receive feedback, and apply their knowledge in practical situations.

Another key objective is to identify the **essential features that contribute to effective learning** in digital environments. These include factors such as structured content, interactive activities, feedback mechanisms, and user engagement strategies that promote understanding rather than memorization.

Furthermore, the research aims to develop a **Pedagogical Evaluation Framework**, which can be used as a standardized method to measure the educational quality of learning applications. This framework will help in systematically evaluating and comparing different apps based on their learning effectiveness.

Finally, the objective is to classify learning applications into different categories, such as high-quality educational apps and low-quality engagement-based apps, based on their pedagogical alignment and overall performance.

Tools/Technologies

In this research project, a combination of basic and accessible tools has been used to ensure a smooth and structured evaluation of online learning applications.

The selection of tools has been made keeping in mind simplicity, efficiency, and the ability to handle different types of data effectively. Since the project is research-oriented rather than purely development-based, the focus is mainly on data collection, analysis, and evaluation rather than heavy software implementation.

The primary programming language used in this project is **Python**, which is widely known for its simplicity and flexibility. Python is chosen because it provides a large number of libraries that can be used for data handling, analysis, and basic processing tasks. It allows the user to organize and manipulate data efficiently, making it suitable for research-based projects. Additionally, Python is beginner-friendly, which makes it easier to implement and understand different parts of the project without requiring complex coding knowledge.

For data storage and organization, **Microsoft Excel or spreadsheet software** has been used. This tool plays an important role in maintaining a structured dataset of selected learning applications. All relevant information such as app features, interaction types, engagement level, and evaluation scores is recorded in a tabular format. This makes it easier to compare different applications and analyze patterns in a systematic way. Excel also helps in performing basic calculations and organizing large amounts of data in a clear and readable format.

The main source of data collection for this project includes platforms such as the **Google Play Store and Apple App Store**. These platforms provide access to a wide range of online learning applications along with detailed descriptions, user reviews, and ratings. By exploring these platforms, different applications can be selected based on their relevance, popularity, and target audience. This helps in creating a diverse dataset for evaluation.

In addition to this, **research papers and academic articles** have been used as supporting resources to understand the theoretical aspects of the project. Concepts such as Bloom's Taxonomy and Evidence-Based Learning are studied through reliable academic sources to build a strong foundation for the evaluation framework. These resources ensure that the project is based on scientifically proven educational principles rather than assumptions.

Overall, the combination of Python, spreadsheet tools, app store data, and academic research provides a well-balanced approach for conducting this study. These tools help in making the research process more organized, reliable, and easy to understand.

METHODOLOGY

The methodology of this project follows a systematic and structured approach to ensure that the evaluation of online learning applications is accurate, consistent, and reliable. The process is designed in such a way that it combines both theoretical understanding and practical analysis, allowing a balanced evaluation of different applications.

The first stage of the methodology involves conducting a detailed **literature review**, where various research papers, articles, and educational theories are studied. This step helps in understanding how learning actually takes place and what factors contribute to effective education. Concepts such as Bloom's Taxonomy and Evidence-Based Learning are analyzed to build a strong theoretical foundation. This ensures that the project is not based on assumptions but on proven learning principles.

After gaining theoretical knowledge, the next stage focuses on the **design of the evaluation framework**. In this phase, a structured Pedagogical Evaluation Framework is developed, which includes key parameters such as engagement level, interaction quality, and cognitive development. These parameters are carefully selected to measure whether an application supports meaningful learning or just focuses on entertainment. Each parameter is defined clearly so that it can be applied uniformly across all selected applications.

The third stage is **data collection**, where a set of online learning applications is selected from platforms like the Google Play Store and Apple App Store. The selection is done based on factors such as popularity, relevance to the target age group, and type of content offered. Detailed information about each application, including its features, learning activities, and interaction mechanisms, is collected and organized in a structured format.

Once the data is collected, the next step is the **evaluation of applications**. In this stage, each application is analyzed using the developed framework. The evaluation focuses on how effectively the app promotes learning, whether it encourages user participation, and how well it supports different levels of cognitive development. Observations are recorded carefully to maintain consistency in evaluation.

The fifth stage involves **scoring and analysis**, where each application is assigned scores based on its performance in different parameters. These scores are then compared to identify patterns, strengths, and weaknesses among different applications. This helps in understanding which applications are more effective and which ones require improvement.

Finally, the results are used for **classification of applications**, where apps are categorized into different groups based on their overall performance. This classification helps in clearly identifying high-quality educational apps and those that are mainly focused on engagement without providing real learning benefits.

To make the methodology clearer and easier to understand, the overall process can be summarized in the following steps:

1. Study existing research and educational theories to understand learning principles.
2. Design a structured evaluation framework based on Bloom's Taxonomy and Evidence-Based Learning.
3. Select relevant online learning applications from app stores.
4. Collect and organize data related to app features and learning activities.
5. Evaluate each application using predefined parameters.
6. Assign scores based on performance in different evaluation criteria.
7. Analyze and compare the results to identify patterns.
8. Classify applications based on their educational effectiveness.

Overall, this methodology ensures that the evaluation process is systematic, transparent, and based on well-defined criteria, making the results reliable and meaningful.

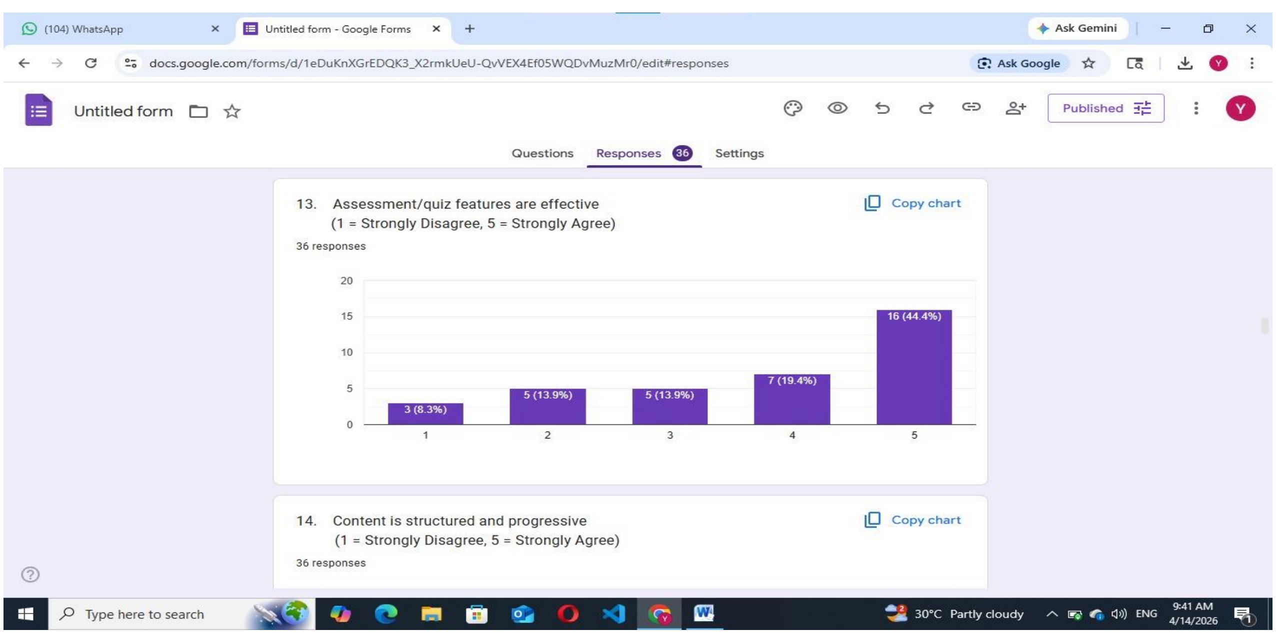
FLOW CHART:

START
↓
Literature Review
(Study Bloom’s Taxonomy & EBL)
↓
Framework Design
(Create Evaluation Criteria)
↓
App Selection
(Choose Learning Applications)
↓
Data Collection
(App Features, Interaction, Activities)
↓
App Evaluation
(Apply Framework)
↓
Scoring System
(Assign Scores Based on Criteria)
↓
Analysis
(Compare Results)
↓
Classification
(High Quality / Low Quality Apps)
↓
Result & Conclusion
↓
END

GOOGLE FORM:

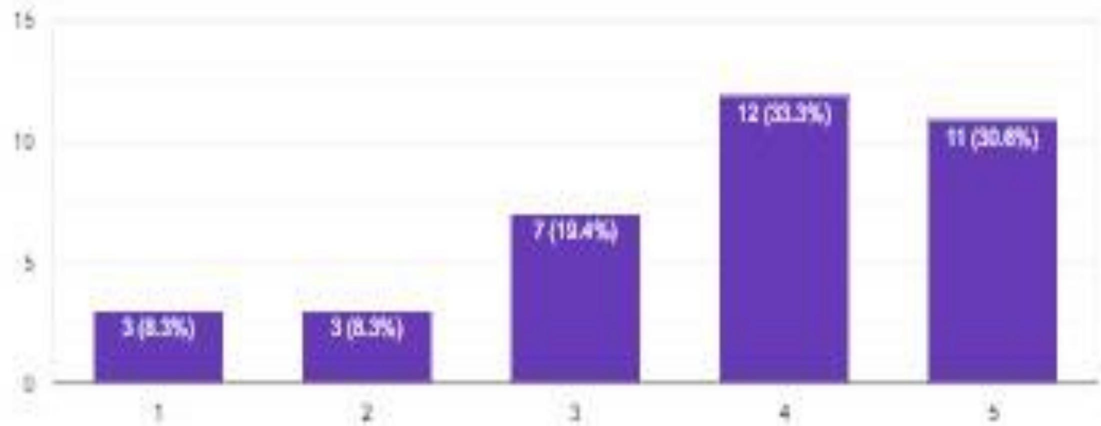
https://docs.google.com/forms/d/e/1FAIpQLSfRF8pMYblp1PVKx2EyKBXxiUK5HrJ_S23jQgWGLZ18i-kNJQ/viewform?usp=publish-editor

GRAPH ANALYSIS:



14. Content is structured and progressive
(1 = Strongly Disagree, 5 = Strongly Agree)

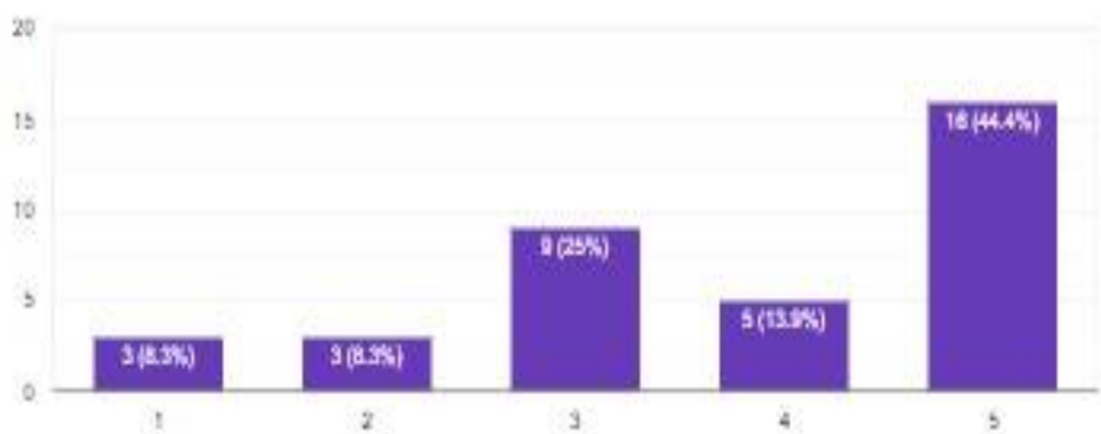
36 responses



SECTION 5: ENGAGEMENT vs ENTERTAINMENT

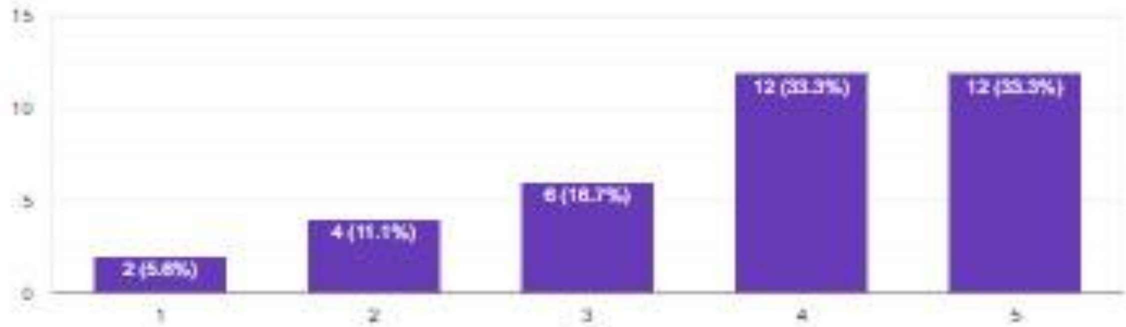
15. The app is engaging for children
(1 = Strongly Disagree, 5 = Strongly Agree)

36 responses



10. The app supports higher-order thinking (analysis, creativity)
(1 = Strongly Disagree, 5 = Strongly Agree)

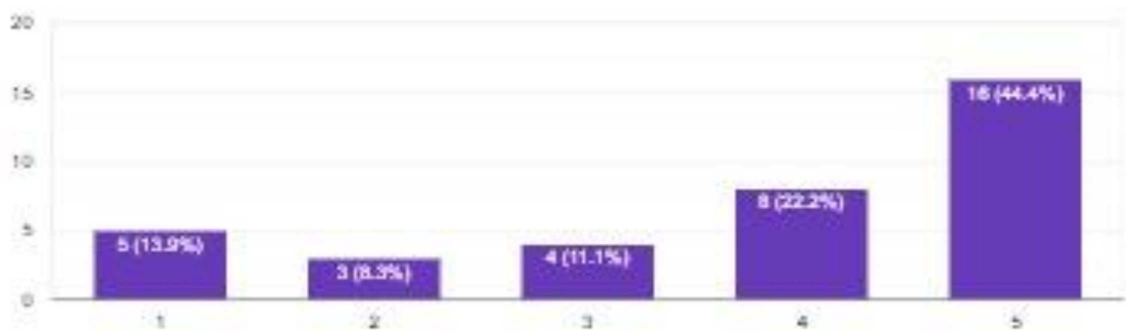
36 responses



SECTION 4: PEDAGOGICAL ALIGNMENT

11. Learning objectives are clearly defined in the app
(1 = Strongly Disagree, 5 = Strongly Agree)

36 responses



12. Activities match the learning goals
(1 = Strongly Disagree, 5 = Strongly Agree)

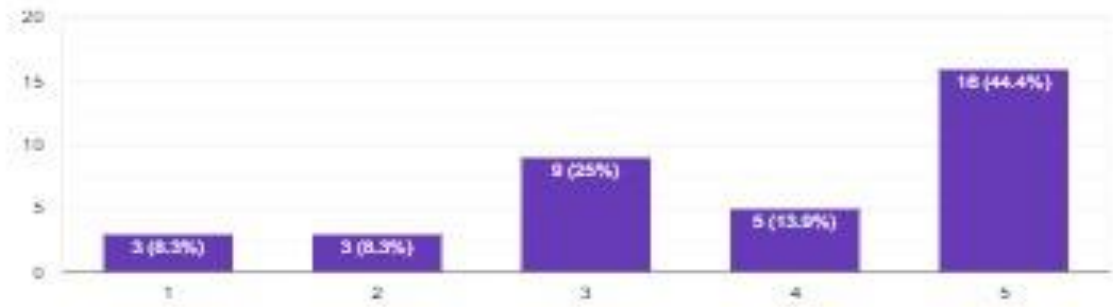
36 responses

SECTION 5: ENGAGEMENT vs ENTERTAINMENT

15. The app is engaging for children
(1 = Strongly Disagree, 5 = Strongly Agree)

Copy chart

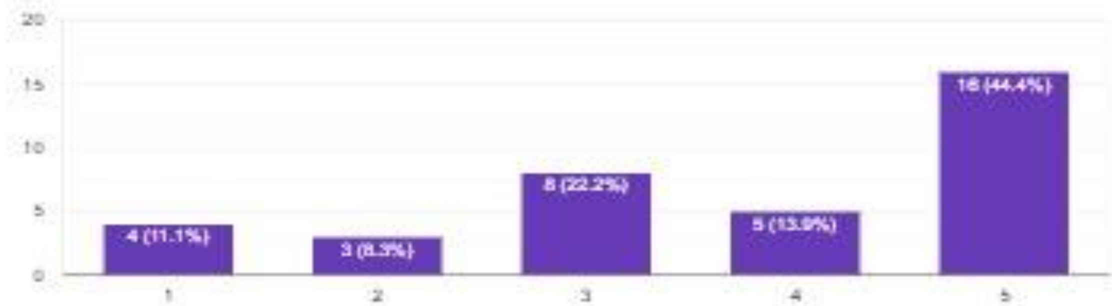
36 responses



16. The app is more entertaining than educational
(1 = Strongly Disagree, 5 = Strongly Agree)

Copy chart

36 responses



17. Animations help in learning
(1 = Strongly Disagree, 5 = Strongly Agree)

Copy chart

36 responses

REFERENCES

Bloom, B. S. (1956). *Taxonomy of Educational Objectives: The Classification of Educational Goals*. New York: Longmans, Green and Co.

Anderson, L. W., & Krathwohl, D. R. (2001). *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*. New York: Longman.

Mayer, R. E. (2009). *Multimedia Learning* (2nd ed.). Cambridge University Press.

Clark, R. C., & Mayer, R. E. (2016). *E-Learning and the Science of Instruction: Proven Guidelines for Consumers and Designers of Multimedia Learning*. Wiley.

Hattie, J. (2009). *Visible Learning: A Synthesis of Over 800 Meta-Analyses Relating to Achievement*. Routledge.

Traxler, J. (2007). "Defining, Discussing and Evaluating Mobile Learning." *The International Review of Research in Open and Distributed Learning*, 8(2).

UNESCO. (2021). *Digital Learning and Transformation of Education*. UNESCO Report.

1. Google: https://www.google.com
2. Google Chrome: https://www.google.com/chrome/
3. YouTube: https://www.youtube.com
4. Wikipedia: https://www.wikipedia.org
5. Safari: https://www.apple.com/safari/
6. Google Play Store: https://play.google.com
7. Apple App Store: https://www.apple.com/app-store/
8. Google Form: https://docs.google.com/forms/d/e/1FAIpQLSfRF8pMYblp1PVKx2EyKBXxiUK5HrJ_S23jQgWGLZ18i-kNJQ/viewform?usp=publish-editor